# Beyond Dehazing: Learning Intrinsic Hazy Robustness for Aerial Object Detection

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***Abstract*— Accurate object detection in aerial imagery is crucial across numerous applications. However, haze can significantly degrade the performance of normal detectors, presenting a substantial obstacle in real-world scenarios. Previous solutions often resort to image dehazing as a pre-processing step to enhance image quality for subsequent detection. Despite being logically intuitive, their performance is limited due to the inherent objective mismatch between low-level image restoration tasks and high-level object detection tasks. In this article, we present haze-robust aerial object detection (HRAOD) to directly enhance detection robustness under hazy conditions. HRAOD constructs a clean-to-hazy distillation framework, enabling the detector to "see through haze," without relying on the explicit image dehazing process. To address the challenge of extracting informative hazy features from blurry and low-contrast hazy images, we introduce a gradient-guided feature imitation method to emphasize the desired objects. Moreover, recognizing that different regions suffer from varying degradation degrees and pose distinct detection difficulties, we further propose a degradation-weighted response distillation method to mimic the normal predictions according to the degradation pattern adaptively. Due to the scarcity of hazy aerial data, we curate two remote sensing hazy aerial datasets, namely DOTA-Haze and SODA-A-Haze, and one drone hazy aerial dataset, DroneVehicle-Haze, for simulation. Extensive experimental results demonstrate the superiority of our method. Specifically, our HRAOD outperforms the state-of-the-art "dehaze + detect" method by 13.1 points in mAP on the DOTA-Haze dataset without incurring additional inference costs. HRAOD also performs favorably against other methods on real-world hazy scenes.**

***Index Terms*— Aerial images, hazy conditions, knowledge distillation, object detection.**

## I. Introduction

AIMING at identifying and localizing ground objects from aerial images, aerial object detection is a pivotal task in remote sensing and has been widely used in various domains, such as environmental monitoring [1], urban planning [2], and defense surveillance [3], [4]. Compared to general object detection, aerial object detection is often considered more challenging, primarily due to the densely packed, arbitrarily oriented, and small-scale nature of aerial objects. To tackle the above challenges, many methods have emerged in recent years and have achieved remarkable performance [5], [6], [7], [8], [9], [10]. Despite significant advancements in this field, most of the current methods train and evaluate their models on clean and high-quality images, neglecting the fact that real-world detection often encounters adverse weather conditions and degraded image quality, which can significantly impact performance. For aerial images, one of the most common image degradation factors is haze. Haze not only blurs object edges, increasing localization difficulty, but also homogenizes the features of the background and objects, making recognition more challenging [11], [12], [13]. Detectors pre-trained on normal images struggle to accurately detect aerial objects in hazy scenes, which limits their applicability in real-world scenarios.

To boost the detection robustness against hazy conditions, one straightforward solution is training detectors on the hazy dataset, thereby improving the model's adaptability to haze. However, due to degradation caused by haze, the model often struggles to effectively extract useful features. To mitigate the impact of haze, mainstream methods [14], [15], [16], [17] typically adopt a two-step approach. As shown in Fig. 1(b), they first attempt to obtain haze-free images using image dehazing algorithms [18], [19], and then feed these reconstructed images into a pre-trained detector for object detection. Some studies also propose to jointly optimize the dehazing and detection networks [20], [21], [22]. Despite the intuitive appeal and logical consistency of the "dehazing-and-detection" framework, achieving ideal results with this approach is often challenging due to the inherent objective mismatch between these two tasks. As a low-level vision task, image dehazing aims at estimating the hazy degradation degree and conducting pixel-level reconstruction. Conversely, object detection, as a high-level vision task, focuses less on pixel-level details and more on holistic semantic information for localization and recognition. As a result, improved image quality does not necessarily equate to better detection results, and sometimes it may even degrade detection performance. In addition to potential objective conflicts, introducing image dehazing as a pre-processing step also incurs additional computational costs, which further limits its practical utility.

Considering that image dehazing steps typically do not reliably and consistently enhance the performance of subsequent detection tasks, in this article, we depart from the

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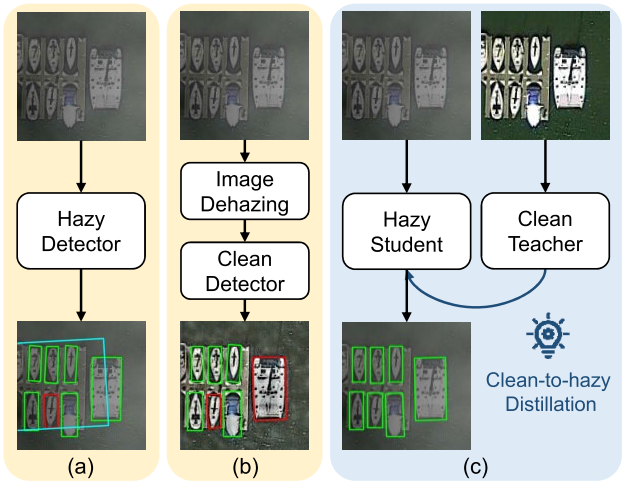



Fig. 1. Framework and performance comparison of different detection methods against hazy conditions. (a) Direct detect: the model trained on the hazy dataset is used for detection. (b) Dehaze + detect: introducing the image dehazing methods as a pre-processing step to obtain clean images, and then feeding these reconstructed images into the clean detector. (c) Distillation (ours): we empower the detector's intrinsic robustness to haze through clean-to-hazy knowledge distillation. The green, blue, and red boxes denote true positive, false positive, and false negative predictions.

traditional "dehazing-and-detection" framework and develop haze-robust aerial object detection (HRAOD), which learns hazy robust detection via knowledge distillation, as depicted in Fig. 1(c). Different from conventional knowledge distillation where a more powerful teacher guides the student network to extract more representative features from the same scene, our HRAOD constructs a clean-to-hazy distillation scheme tailored for detection under hazy conditions. Specifically, the teacher network, trained on clean aerial images, serves as a guide for the student network to learn how to extract degradation-resilient features and accurately identify objects from degraded hazy images. This clean-to-hazy distillation enables the detector itself to adapt to the degradation process. To promote the transmission of clean knowledge and assist detectors in overcoming degraded environments, we further propose two distillation strategies. As depicted in Fig. 2, hazy images typically exhibit low contrast. The edges and details in hazy images are less pronounced compared to those in clean images. These characteristics pose challenges for the backbone network in extracting representative features, consequently affecting subsequent classification and recognition. To emphasize the semantic information, we propose a gradient-guided feature imitation strategy to highlight potential objects in hazy features. Further, considering the nonuniform distribution of haze, we propose a degradation-weighted response distillation strategy to encourage the student network to adaptively imitate the predictions of the teacher network, thereby enhancing the detector's capabilities in accurately localizing and recognizing objects in hazy scenes. In summary, our contributions are as follows.

1) We propose HRAOD, the first knowledge distillation framework for enhancing detection robustness in hazy aerial scenes. HRAOD achieves significant performance improvement under hazy conditions without sacrificing efficiency.

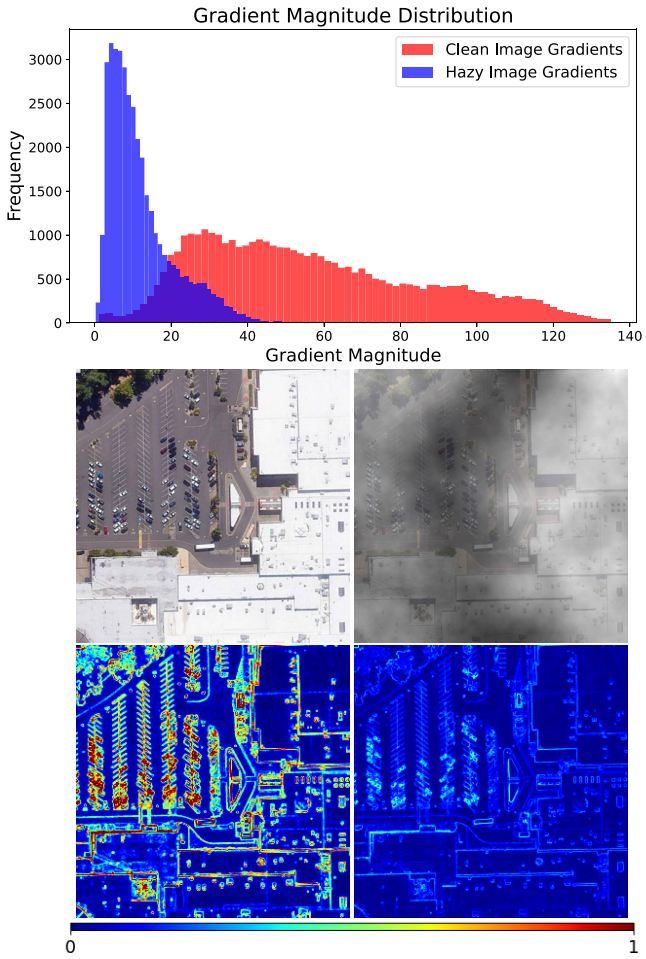



Fig. 2. (Top) Statistical distribution of gradient magnitudes for clean dataset (SODA-A) and hazy dataset (SODA-A-Haze). (Bottom) Gradient magnitudes heatmap for clean and hazy images in the same scene. Haze not only homogenizes the features of the background and objects but also blurs the edges of objects, reducing contrast.

2) We design two distillation strategies: gradient-guided feature imitation and degradation-weighted response distillation, to encourage the clean-to-hazy knowledge transfer. These two strategies empower the detector to "see through haze."
3) We introduce two remote sensing aerial hazy datasets (DOTA-Haze and SODA-A-Haze) for experiments. Additionally, we conduct experiments on the DroneVehicle dataset, which contains some real-world hazy images, and we specifically selected these images to constitute a test set. Our HRAOD outperforms other methods on both synthetic and real-world hazy images.

The remainder of this article is structured as follows. Section II offers a comprehensive review of aerial object detection, object detection under hazy conditions, and knowledge distillation. Then, Section III provides a detailed introduction to our proposed HRAOD. Extensive experiments are conducted and presented in Section IV. In Section V, we analyze the impact of haze on different objects and how our method works on clean dataset. We also discuss the impact of haze intensity in this section. Finally, we draw a conclusion in Section VI.

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## II. Related Works

### A. Aerial Object Detection

With the rapid development of imaging technology, aerial object detection has become a hot research topic in recent years. Compared to general object detection tasks, aerial object detection is more challenging mainly because objects in aerial images are typically smaller and have arbitrary orientations. Early methods [5], [6], [23] often resorted to massive rotated anchors with different sizes and angles to generate enough proposals, which required significant computational resources. To overcome this limitation, several methods were proposed to convert horizontal region of interests (RoIs) to rotated RoIs for efficiency [7], [24], [25]. RoI Transformer [24] and ReDet [7] designed learnable modules to transform horizontal RoIs into rotated RoIs. In DoDet [25], the oriented proposals were directly generated by the network, thereby circumventing the need for complex RoI operations. Although these approaches have achieved certain effectiveness, they often rely on rotation angle representation, resulting in loss discontinuity and regression inconsistency. To address this issue, Oriented RepPoints [8] proposed using adaptive points representation for diverse orientation shapes and poses, thereby better capturing the geometric structure of aerial objects.

Due to the presence of small and densely arranged objects in aerial images, some methods focus on enhancing the representation capability of deep features [6], [26], [27], [28], [29]. $R^3$Det [26] and COF-Net [29] proposed feature refinement strategies to address feature misalignment. In [27] and [28], the non-local attention mechanism was introduced to aggregate global contextual information and mitigate background noise. Some other works also focus on designing more effective loss functions to supervise the detector [30], [31], [32], [33]. GWD [30] and KLD [31] transformed rotated boxes into a 2-D Gaussian distribution and designed distance functions to model the deviation between two oriented bounding boxes (OBBs). To further leverage the advantages of Gaussian-based loss, KFIoU was proposed in [32], which better fit the skew intersection over union loss and did not require hyperparameter tuning. Recently, ARUBA [9] developed an architecture-agnostic balance to handle the size imbalance of aerial images. MENet [34] proposed to enhance the detection performance of tiny object from both the instance and feature levels. Despite the emergence of many aerial object detection works, most of them focus on aerial images under normal weather conditions, which overlooks the image degradation in real-world scenarios. In the presence of common degradation factors such as rain and haze in aerial images, the performance of these algorithms can significantly degrade. Tailoring aerial detection algorithms for adverse conditions holds crucial practical significance and demands urgent attention.

### B. Object Detection Under Hazy Condition

Haze is one of the common degradation factors in object detection. Due to the reduction in contrast between potential objects and the context, as well as the blurring of objects' edges, hazy scenes simultaneously increase the difficulty of object recognition and localization. In recent years, several efforts have been proposed to increase the detection accuracy under hazy conditions. AOD-Net [14] proposed a lightweight end-to-end network for image dehazing and verified its contribution to object detection. In uDnD [15], the haze density prior was leveraged to provide rich domain knowledge for both the dehazing and detection tasks. In DSNet [20], two subnets were designed and trained end-to-end jointly for simultaneously enhancing visibility and detecting objects. Considering the frequency characteristic of degraded images, DENet [16] designed distinct modules to enhance different frequency components of images separately, aiming to better enhance structural features and thus improve detection performance. For efficiency demand, YOLOv5-Fog [17] presented a fast detection algorithm for automatic driving perception under hazy conditions. Recently, IA-YOLO [21] proposed a white-box differentiable module for dehazing and directly supervised the dehazing-and-detection network by detection loss. In BAD-Net [22], a two-branch structure with an interval iterative training strategy was proposed to achieve detection-friendly dehazing. Despite significant progress, most of these methods still involve explicit image dehazing processes, which may not align optimally with the objectives of detection tasks. Improved image quality does not necessarily translate to better detection results, leading to suboptimal or even worse outcomes. In contrast to existing methods, our proposed HRAOD adopts knowledge distillation to enable the detector to "see through haze" without relying on an explicit image dehazing process.

### C. Knowledge Distillation for Object Detection

Leveraging powerful teacher models to guide simple student models, knowledge distillation is initially aimed at maintaining network performance while compressing model size. According to the distillation information, current models can be categorized into three types: feature-based distillation [35], [36], [37], [38], [39] aims at aligning the feature maps of teacher model and student model for powerful representations. Response-based distillation [37], [40], [41], [42] uses the final predictions to guide the optimization process of student model for precise outputs. Relation-based distillation [43], [44], [45], [46] aims at transferring the interactive information between different instances, layers, and positions to enhance the student's generalization capability.

In recent years, knowledge distillation has also been considered for addressing object detection under adverse cases. Li et al. [47] and Hnewa et al. [48] introduced knowledge distillation to enrich information and surpass noise under low-light conditions. In [49] and [50], the high-quality cross-domain knowledge was utilized to enhance the detection performance in low-quality target domains. Most recently, Huang et al. [51] developed a distillation framework for 3D object detection under adverse weather. CMHRD [52] proposes to distill information from RGB images to thermal infrared images, thereby addressing the issues of low resolution and limited information in thermal infrared images. Despite the progress made by current methods, none of the existing works have explored whether or how knowledge

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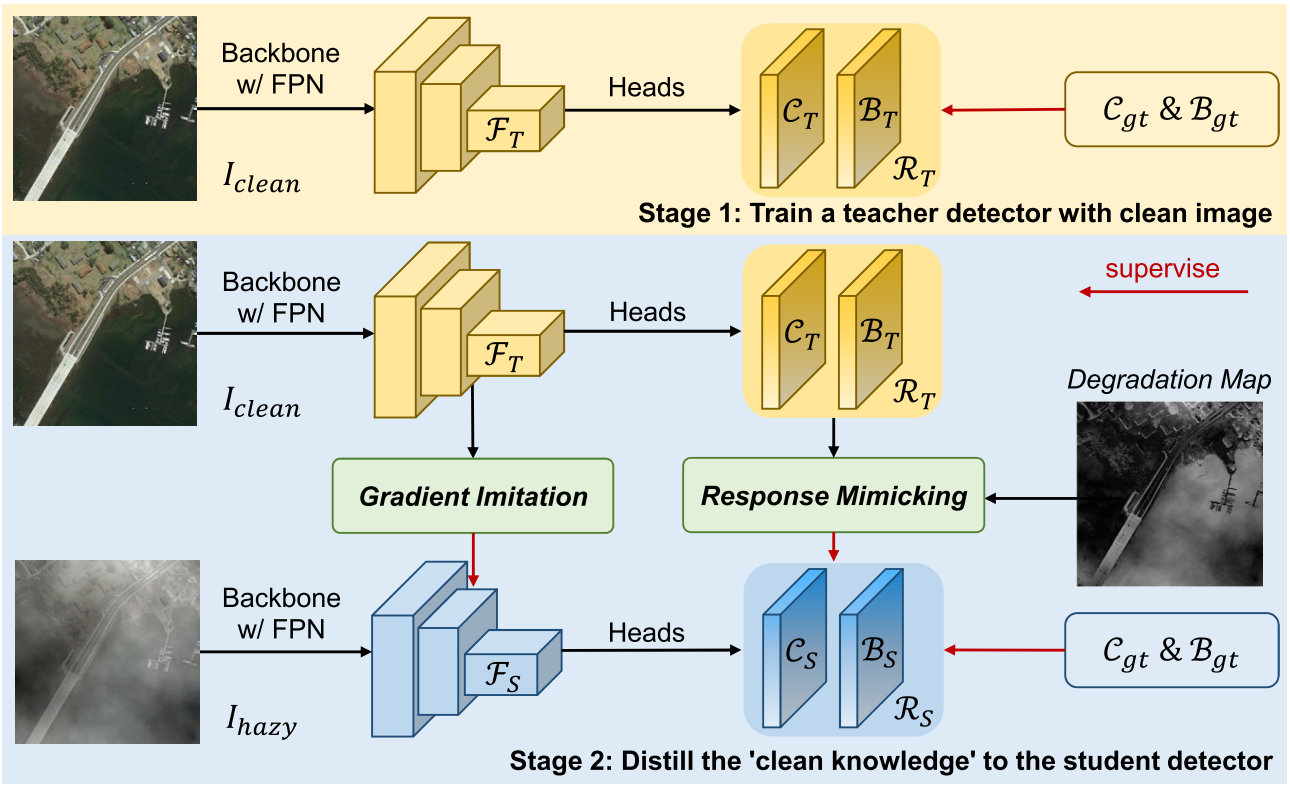



Fig. 3. Overall framework of our proposed HRAOD. We construct a clean-to-hazy knowledge distillation paradigm with two distillation strategies to enhance the detection robustness under hazy scenes. $\mathcal{F}$ refers to the feature extracted by FPN. $\mathcal{C}$ and $\mathcal{T}$ refer to the output response maps of the classification head and regression head, and we use $\mathcal{R}$ to represent the response set of detection heads.

distillation can benefit object detection in hazy conditions, a common degradation scenario in aerial scenes [11].

## III. Methods

For object detection under hazy conditions, conventional methods often follow the "dehaze + detection" paradigm, whose effectiveness is highly dependent on the dehazing performance. However, due to the inherent objectives mismatch between image restoration tasks and detection tasks, these methods can only obtain suboptimal, and sometimes even worse detection performance. This limitation constrains the practical application of object detection techniques in real-world scenarios. To tackle this issue, we propose a novel framework for learning hazy robust detectors via knowledge distillation, termed HRAOD. A teacher network trained with clean images will help the student network more accurately locate and recognize hazy objects.

### A. Overall Framework

The overall framework of our proposed HRAOD is illustrated in Fig. 3 and comprises two stages. In stage 1, we first train a teacher detector $D_T$ with clean images $I_{\text{clean}} \in \mathbb{R}^{H\times W\times 3}$ as inputs, where $H$ and $W$ refer to the height and width of the image. The loss function of $D_T$ can be formulated as follows:

$$L_T = L_{\text{cls}}(\mathcal{C}_T, \mathcal{C}_{\text{gt}}) + \alpha L_{\text{bbox}}(\mathcal{B}_T, \mathcal{B}_{\text{gt}}) \tag{1}$$

where $L_{\text{cls}}$ is the classification loss, $L_{\text{bbox}}$ is the bounding box regression loss, and $\alpha$ is the weight factor of $L_{\text{bbox}}$. $\mathcal{C}_T$ and $\mathcal{B}_T$ denote the output response of the classification head and regression head. $\mathcal{C}_{\text{gt}}$ and $\mathcal{B}_{\text{gt}}$ refer to the classification and localization labels.

In stage 2, we train the student detector $D_S$ with hazy images $I_{\text{hazy}} \in \mathbb{R}^{H\times W\times 3}$, and input the corresponding paired clean images $I_{\text{clean}}$ into $D_T$ for inference. It is worth mentioning that we simulate the hazy images from clean ones since currently there is no hazy aerial data specifically for object detection. Then, we transfer the informative "clean knowledge" to the student detector $D_S$, thereby enhancing the detection accuracy and robustness under hazy conditions. Specifically, considering the degradation pattern of hazy images, we tailored two distillation loss functions to help the student network $D_S$ better "see through haze." The loss function of $D_S$ can be expressed as follows:

$$\begin{aligned} L_S = {} & L_{\text{cls}}(\mathcal{C}_S, \mathcal{C}_{\text{gt}}) + \alpha L_{\text{bbox}}(\mathcal{B}_S, \mathcal{B}_{\text{gt}}) \\ & + \beta L_{\text{grad}}(\mathcal{F}_S, \mathcal{F}_T) + \gamma L_{\text{degra}}(\mathcal{R}_S, \mathcal{R}_T, M) \end{aligned} \tag{2}$$

where $L_{\text{grad}}$ refers to the proposed gradient-guided feature imitation loss with weight $\beta$, aiming at enhancing the objective-background contrast and mitigating feature ambiguity. $L_{\text{degra}}$ refers to the proposed degradation-weighted response loss with weight $\gamma$, for encouraging the $D_S$ to produce more accurate outputs. $\mathcal{G}$ denotes the gradient representation of deep feature maps, and $\mathcal{R}$ represents the output response of detection head. $M$ is the estimated degradation map for controlling the distillation weights of different regions. Following we will give a detailed description of our method.

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### B. Gradient-Guided Feature Imitation

Under the hazy conditions, the aerial images are severely degraded, resulting in a reduction in contrast between the foreground and the background when compared to the clean image. This diminished contrast poses a challenge for the network in effectively extracting valuable features. Furthermore, haze often induces a blurring effect on the entire image, leading to a loss of semantic clarity in the feature map. The blurring effect makes it difficult to maintain sharp and well-defined semantic boundaries between objects, thereby increasing the localization difficulty. To address the above challenges, one intuitive solution is adopting feature imitation to boost the feature quality of the student detector. However, due to the excessive gap between clean features and haze features, the student network is unable to grasp the focus of feature imitation. Keeping these challenges in mind, we propose a simple but effective gradient-guided feature imitation method to highlight the desired objects in haze scenes.

Specifically, unlike traditional feature imitation methods that force the student model's features to approach the teacher's, we propose to conduct distillation based on the gradient information, which carries more semantic information and reveals the outline of objects. For the teacher features $\mathcal{F}_T \in \mathbb{R}^{H_f \times W_f \times C}$ and the student features $\mathcal{F}_S \in \mathbb{R}^{H_f \times W_f \times C}$ with spatial size $H_f \times W_f$ and $C$ channels, we first adopt Sobel operator to extract their gradients. Then, considering the channel alignment problem in knowledge distillation, we further compress their gradients across channel dimensions to obtain the gradient representations $\mathcal{G}_S$ and $\mathcal{G}_T$. The above procedure can be formulated as follows:

$$\mathcal{G}_S = \frac{1}{C}\sum_{c=1}^{C}\sqrt{S_x(\mathcal{F}_S)^2 + S_y(\mathcal{F}_S)^2} \tag{3}$$

$$\mathcal{G}_T = \frac{1}{C}\sum_{c=1}^{C}\sqrt{S_x(\mathcal{F}_T)^2 + S_y(\mathcal{F}_T)^2} \tag{4}$$

where $S_x$ and $S_y$ refer to the Sobel operators in the horizontal and vertical directions, respectively. After obtaining $\mathcal{G}_T$ and $\mathcal{G}_S$, we conduct gradient imitation to highlight the edge information of the feature maps. The proposed gradient-guided imitation loss $L_{\text{grad}}$ is expressed as follows:

$$L_{\text{grad}} = \|\mathcal{G}_T - \mathcal{G}_S\|_1. \tag{5}$$

By imposing imitation constraints on the gradient representations, we enhance the discrimination between objects and backgrounds in haze features, thereby reducing the difficulty of localization and classification. Additionally, conducting imitation on the gradient information rather than the entire feature map serves to alleviate the complexity of distillation learning. Consequently, the student detector can better grasp the focal points of distillation learning, facilitating a more effective transfer of the information learned by the teacher network.

### C. Degradation-Weighted Response Distillation

In aerial images, the distribution of haze is usually nonuniform. Objects in different regions are typically obscured by haze with varying concentrations, resulting in varying degrees of difficulty in localization and recognition. While light haze may not significantly impact detection performance, the presence of heavy haze can severely obscure objects, resulting in missed or false detection. To address this challenge, we propose a degradation-weighted response distillation strategy, where the response distillation process is adaptively weighted by the haze intensity of different regions. This strategy enables the teacher network to allocate more attention to the distillation learning of severely degraded regions, thereby enhancing the detection robustness against harsh degradation.

**Algorithm 1** Degradation Map Estimation

**Data**: Paired clean image $I_{clean}$ and hazy image $I_{hazy}$.
1 Calculate the absolute pixel-level difference of the paired images: $I_d = abs(I_{clean} - I_{hazy})$.
2 Average along channel dimension: $I_{avg} = \frac{1}{3}\sum_{c=1}^{3} I_d$.
3 Min-Max Normalization: $M = \frac{((I_{avg} - \min(I_{avg})))}{((\max(I_{avg}) - \min(I_{avg})))}$.
**Output**: The estimate pixel-level degradation map $M$.

Specifically, given the paired clean image $I_{\text{clean}}$ and hazy image $I_{\text{hazy}}$, we estimate the degradation map by Algorithm 1. First, we calculate the absolute pixel-level difference between the paired images. Then, the difference is further averaged along the channel dimension. Last, the min-max normalization is conducted to ensure numerical stability. The obtained degradation estimation map $M \in \mathbb{R}^{H \times W}$ quantifies the degradation level in different regions of a hazy scene, and we utilize it to determine the learning priority for the student network during the distillation. Given the output responses $\mathcal{R}_T = \{\mathcal{C}_T, \mathcal{B}_T\}$ and $\mathcal{R}_S = \{\mathcal{C}_S, \mathcal{B}_S\}$, the degradation-weighted response distillation $L_{\text{degra}}$ can be formulated as

$$L_{\text{degra}} = M \odot L_{\text{kd}}(\mathcal{C}_S, \mathcal{C}_T) + M \odot L_{\text{ld}}(\mathcal{B}_S, \mathcal{B}_T) \tag{6}$$

where $L_{\text{kd}}$ refers to the commonly used KD loss [53] for distilling the classification knowledge. For the localization knowledge distillation $L_{\text{ld}}$, we first convert the standard bounding box representation into probability distribution form [41], which carries abundant position information. Then, we formulate the localization distillation loss as follows:

$$L_{\text{ld}} = L_{\text{KL}}\left(\mathcal{S}\left(\frac{\mathcal{B}_T}{\tau}\right), \mathcal{S}\left(\frac{\mathcal{B}_S}{\tau}\right)\right) \tag{7}$$

where $\mathcal{S}$ is the softmax operation, $\tau$ is the distillation temperature, and $L_{\text{KL}}$ is the KL-divergence loss. By incorporating the degradation estimation map $M$, our method empowers the student detector to adjust its learning focus based on the severity of degradation in different regions, leading to improved performance in hazy scenarios.

## IV. Experiments and Results

### A. Datasets

Currently, neither paired clean-hazy aerial image datasets nor labeled hazy aerial image datasets specifically for object detection are available. Hence, we resort to using simulated hazy aerial data for experiments. We generate two remote

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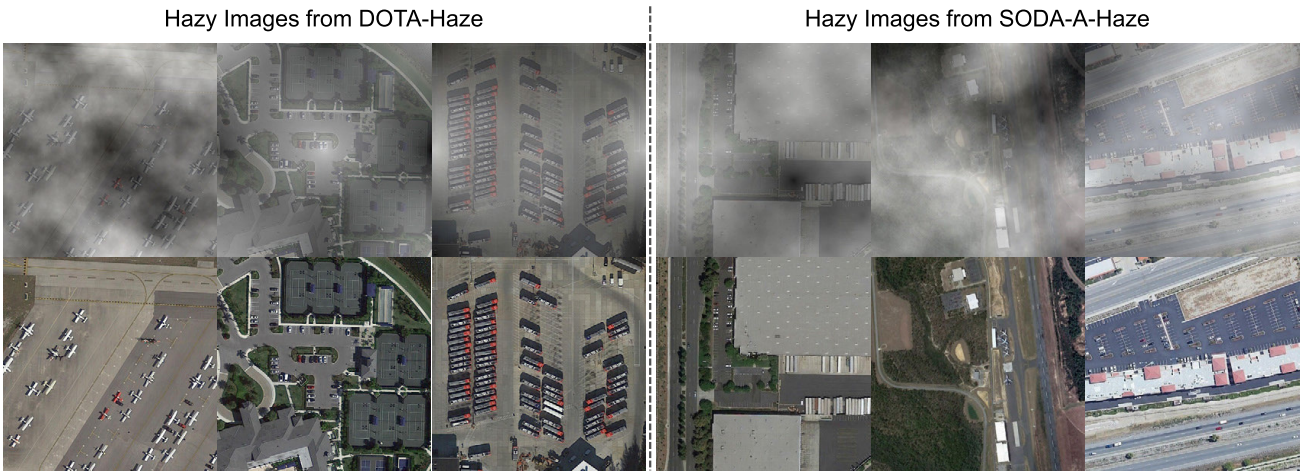



Fig. 4. Some hazy aerial images generated by the ImageNet-C benchmark and their corresponding clean images.

sensing aerial hazy datasets called DOTA-Haze and SODA-A-Haze, and one drone hazy dataset, named DroneVehicle-Haze. DOTA-Haze is based on the DOTA-v1.0 dataset [5], which is a large-scale object detection dataset for aerial images. This dataset consists of 15 object categories, and each object is annotated by the OBB. The categories in the DOTA-Haze dataset are defined as plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Totally 15 749 images with the size of 800 × 800 are used for training, and 5297 images with the size of 800 × 800 are used for testing.

In aerial scenes, the object scale can vary significantly. While large-scale objects can maintain their features and semantic information to some extent in the presence of haze, those small objects are easily obscured by haze. To investigate the impact of haze on small object detection and further verify our effectiveness, we further conduct experiments on SODA-A-Haze dataset. The SODA-A-Haze dataset is based on the newly proposed SODA-A dataset [54], which is a large-scale benchmark for small aerial object detection. This dataset consists of nine categories, and each object is also annotated by the OBB. The categories in the SODA-A-Haze dataset are airplane (AP), helicopter (HC), SV, LV, ship (SH), container (CO), tank (TA), SP, and windmill (WM). Totally 37 980 images with the size of 800 × 800 are used for training, and 30 974 images with the same size are used for testing.

To further validate the robustness and practicability of HRAOD, we extend our experiments to DroneVehicle dataset [55], which encompasses real-world hazy scenes. Specifically, we select 1073 real-world hazy scenes to construct a test set. These scenes are collected from the official training, validation, and testing datasets of DroneVehicle. The remaining 17 309 images from the training datasets are used to curate the DroneVehicle-Haze dataset for model training. The categories in DroneVehicle-Haze dataset are car (CA), truck (TR), bus (BU), van (VA), and freight car (FC).

To generate the corresponding haze dataset, we adopt the ImageNet-C benchmark [56] to generate haze corruption on the aerial image. Specifically, given the clean image $I_{\text{clean}} \in \mathbb{R}^{H\times W\times 3}$, the corresponding hazy image can be generated as follows:

$$I_{\text{hazy}} = (I_{\text{clean}} + c_1 \cdot H_d(H, W, c_2)) \cdot \frac{\max(I_{\text{clean}})}{\max(I_{\text{clean}}) + c_1} \quad (8)$$

where $H_d$ refers to the diamond-square algorithm used to generate the haze corruption. $c_1$ and $c_2$ are the hyperparameters that control the haze intensity. To enhance the diversity of the data and strengthen the model's robustness to different levels of haze, we randomly selected severity parameters provided by ImageNet-C during data generation. Some examples of paired hazy-clean images are shown in Fig. 4.

### B. Experimental Setup

In our implementation, we select the rotated RetinaNet [60] as the baseline detector, where ResNet50 pre-trained on the ImageNet [61] dataset is utilized as the backbone. All input images are resized to 1024 × 1024 before training, and we use the P3–P7 feature maps in FPN for obtaining multiscale information. We follow the schedule_1x training setting in MMrotate [62], where all detection models are trained in 12 epochs. The learning rate is initialized as 0.0025 and is divided by 10 at epochs 8 and 11. All detection models are trained on one NVIDIA GeForce RTX 3090 GPU with a batch size of 4. For the image dehazing methods, we first train these methods on the training sets following their official settings. The clean-hazy image pairs are treated as inputs and targets. Then, we use these image dehazing methods to generate the reconstructed images of the testing sets (the testing set of DOTA-Haze and SODA-A-Haze). To evaluate the detection performance quantitatively, we adopt the widely used mean average precision (mAP) as the evaluation metric.

As for the loss functions and their weights, we adopt FocalLoss [60] as $L_{\text{cls}}$. Since the bounding box representation is modeled by the Gaussian distribution, as in VLR [41]. We simply keep its setting, using GWD loss [30] with weight $\alpha = 5$ as $L_{\text{bbox}}$. For distillation losses, the temperature $\tau$ is set to 10. To make the scale of the distillation loss similar to that of the detection loss, we empirically set the weight factors $\beta$ and $\gamma$ to 10.0 and 3.0, respectively.

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TABLE I
QUANTITATIVE DETECTION COMPARISON ON THE DOTA-HAZE DATASET. BOLD TEXT SHOWS THE BEST RESULTS

| Method | PL | BD | BR | GTF | SV | LV | SH | TC | BC | ST | SBF | RA | HA | SP | HC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Directly Detect:** | | | | | | | | | | | | | | | | |
| Clean Detector | 77.1 | 24.3 | 11.3 | 18.8 | 23.2 | 21.8 | 63.8 | 81.4 | 37.5 | 50.8 | 27.6 | 28.3 | 39.3 | 9.7 | 10.5 | 35.0 |
| Hazy Detector | 87.0 | 58.5 | 27.7 | 39.4 | 64.6 | 61.8 | 78.0 | 90.2 | 52.6 | **60.0** | 44.5 | 52.6 | 54.2 | 52.6 | 45.6 | 58.0 |
| **Dehaze + Detect:** | | | | | | | | | | | | | | | | |
| w/ DCP [57] | 68.4 | 35.1 | 10.7 | 26.4 | 53.6 | 43.7 | 73.7 | 80.8 | 36.6 | 51.2 | 28.5 | 35.5 | 47.2 | 37.6 | 14.5 | 42.9 |
| w/ AOD-Net [14] | 62.8 | 25.6 | 9.5 | 19.5 | 23.9 | 21.6 | 59.6 | 80.6 | 31.5 | 48.9 | 25.9 | 24.9 | 38.4 | 13.9 | 3.5 | 32.7 |
| w/ GridDehaze [58] | 80.4 | 47.5 | 16.7 | 23.7 | 53.1 | 48.6 | 75.8 | 88.6 | 49.0 | 58.6 | 31.5 | 47.9 | 38.3 | 26.6 | 28.5 | 47.7 |
| w/ EMPF-Net [59] | 80.5 | 42.0 | 13.9 | 24.7 | 54.5 | 45.4 | 76.4 | 87.7 | 48.4 | 57.4 | 34.7 | 37.2 | 35.2 | 34.5 | 30.6 | 46.9 |
| **Distillation:** | | | | | | | | | | | | | | | | |
| HRAOD (ours) | **88.0** | **64.8** | **28.5** | **49.7** | **66.3** | **66.4** | **78.5** | **90.5** | **54.8** | **60.0** | **45.1** | **55.8** | **59.2** | **53.9** | **49.9** | **60.8** |

Clean Detector is trained on clean images, and directly tested on hazy images. Hazy Detector is trained and tested on hazy images.

### C. Experimental Results on DOTA-Haze

To verify the effectiveness of our proposed HRAOD, we compare our method with several approaches for hazy image object detection, and the quantitative results are shown in Table I. For directly detect, we directly evaluate the detection performance of the baseline model, rotated RetinaNet, in hazy aerial scenes. For dehaze + detect, we select four representative single-image dehazing algorithms as the pre-processing step and use rotated RetinaNet trained on normal aerial images for detection. We select DCP [57], AOD-Net [14], GridDehaze [58], and EMPF-Net [59] as the image dehazing methods, encompassing both classic and state-of-the-art algorithms. These methods are abbreviated as w/ DCP, w/ AOD, w/ GridDehaze, and w/ EMPF-Net in the following tables and figures.

Table I compares the mAP results between all methods. It can be observed that our HRAOD achieves the best detection performance under hazy conditions. Based on these experiment results, some analysis and conclusions are derived as follows.

1) For directly detect methods, it can be observed that the performance of the clean detector significantly deteriorates in the presence of haze. Although training the detector on hazy images (denoted as hazy detector) can improve the results to some extent, the effectiveness of this method still suffers from the localization blurring and classification difficulties brought about by the haze.
2) For dehaze + detect methods, their performance largely depends on the capabilities of the image dehazing networks. While the DCP, GridDehaze, and EMPF-Net can improve the detection performance to some extent, AOD-Net, on the contrary, reduces the detection performance. This result reflects that low-level image restoration tasks do not always contribute to high-level object detection tasks. It is challenging to find a robust image dehazing method for every hazy scene, and it is also difficult to ensure that the dehazed images are helpful for object detection.
3) Different from the mainstream approaches, our HRAOD does not rely on image dehazing as a pre-processing step to enhance the quality of images. Instead, we adopt the knowledge distillation framework, utilizing the information extracted from clean images to guide the feature extraction, localization, and classification procedures of the detector under hazy scenes. Our proposed degradation-weighted response distillation loss strengthens the detector's ability to locate and recognize hazy objects, and our proposed gradient-guided feature imitation loss enhances the semantic clarity of haze features, guiding the network to efficiently extract features from hazy images. As a result, our method outperforms all methods on all scenes. In order to further underscore the superiority of our proposed HRAOD, we provide a qualitative analysis and present visual results in Fig. 5. It is evident from the visualizations that our method effectively locates and recognizes objects in hazy scenes, even when dealing with small and densely arranged instances, which further validates its effectiveness.

### D. Experimental Results on SODA-A-Haze

To investigate the impact of haze on small aerial object detection and further verify our effectiveness, we conduct experiments on the SODA-A-Haze dataset. The experiment settings are kept the same as those on the DOTA-Haze dataset. As demonstrated in Table II, our method consistently improves the detection robustness against haze degradation in the scenario of small object detection. Compared to the top-performing dehaze + detect framework, our approach surpasses it by 16.1 points on mAP. In Fig. 6, we further demonstrate the visualization of detection results. It can be observed that image dehazing methods cannot consistently improve detection performance. In the first three scenarios, these dehazing methods could enhance detection performance to some extent compared with the baseline model. However, in the fourth scene, only AOD-Net produced a slight improvement. Although image dehazing can yield visually clear and appealing results, they do not directly contribute to high-level

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TABLE II
QUANTITATIVE DETECTION COMPARISON ON THE SODA-A-HAZE DATASET. BOLD TEXT SHOWS THE BEST RESULTS

| Method | AP | HC | SV | LV | SH | CO | TA | SP | WM | mAP |
|---|---|---|---|---|---|---|---|---|---|---|
| **Directly Detect:** | | | | | | | | | | |
| Clean Detector | 34.2 | 5.2 | 17.3 | 10.7 | 16.5 | 18.9 | 25.1 | 17.6 | 15.9 | 17.9 |
| Hazy Detector | **85.4** | 37.0 | 69.8 | 31.5 | 63.2 | 37.6 | 69.8 | 81.7 | **60.7** | 59.6 |
| **Dehaze + Detect:** | | | | | | | | | | |
| w/ DCP [57] | 37.9 | 2.9 | 38.3 | 15.1 | 35.5 | 28.2 | 45.5 | 56.1 | 17.8 | 30.8 |
| w/ AOD-Net [14] | 68.2 | 13.8 | 56.6 | 18.9 | 60.5 | 31.9 | 56.5 | 41.6 | 32.7 | 42.3 |
| w/ GridDehaze [58] | 76.9 | 17.5 | 48.8 | 20.2 | 47.8 | 36.7 | 60.8 | 63.7 | 43.9 | 46.2 |
| w/ EMPF-Net [59] | 76.1 | 16.8 | 48.5 | 22.1 | 49.0 | 36.1 | 59.0 | 66.5 | 36.4 | 45.6 |
| **Distillation:** | | | | | | | | | | |
| HRAOD (Ours) | 85.0 | **39.9** | **71.4** | **33.9** | **69.6** | **45.9** | **71.8** | **82.9** | 60.6 | **62.3** |

Clean Detector is trained on clean images, and directly tested on hazy images. Hazy Detector is trained and tested on hazy images.

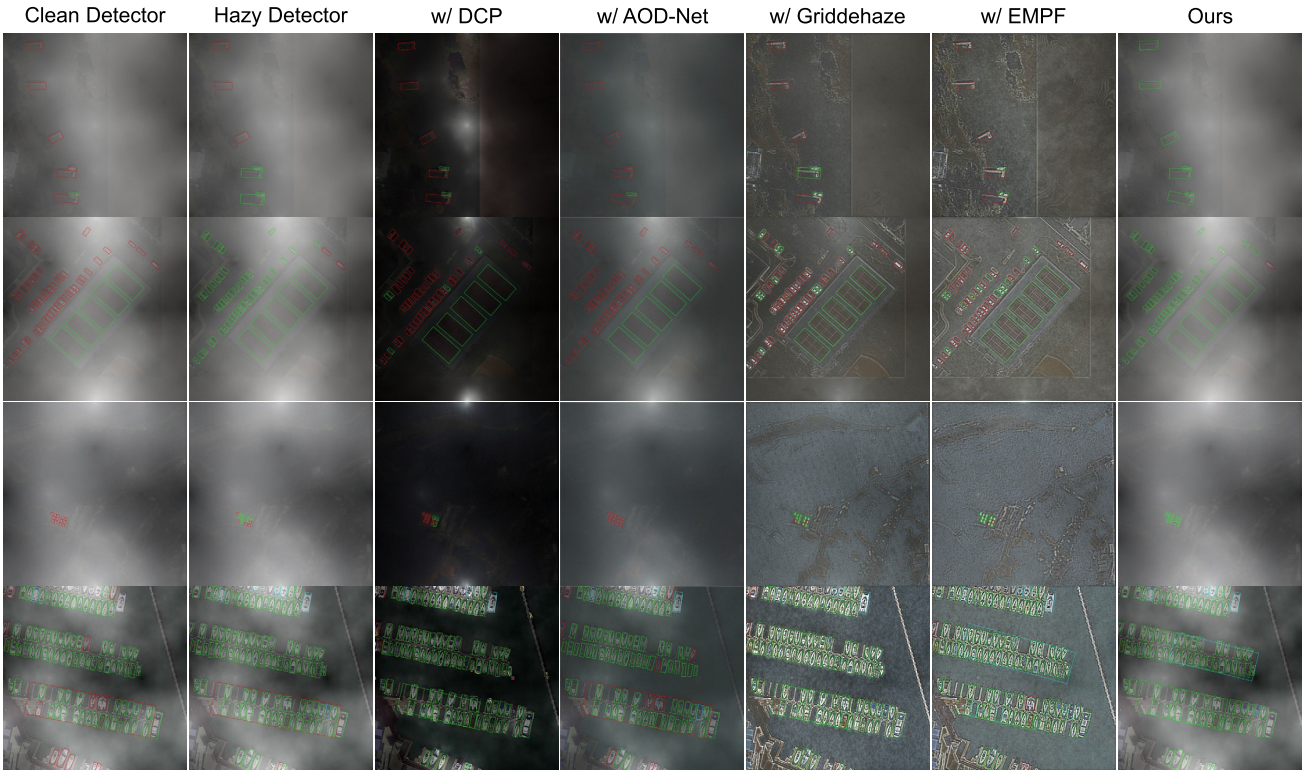



Fig. 5. Qualitative detection comparison on the DOTA-Haze dataset. We use green, blue, and red boxes to denote true positive, false positive, and false negative predictions, respectively.

vision tasks, and sometimes they even generate over-dehazed images that could be harmful to object detection. In contrast to these methods, our HRAOD directly empowers the detector to detect under hazy conditions, leading to fewer missed detections.

### *E. Experimental Results on Real-World Aerial Hazy Scenes*

To further compare the generalization ability of different methods, we evaluate these methods on real-world aerial hazy images. As can be seen from the visual results depicted in Fig. 7, the clean detector struggles to identify objects obscured by haze, leading to unsatisfactory performance. Although the hazy detector detects more targets, it generates many false predictions. As for the dehaze + detect method, although EMPF-Net achieves relatively stable improvement on simulated hazy datasets, it produces severe color distortion and generates over-dehazed results, which is detection-unfriendly. Other dehazing methods also fail to effectively improve detection performance. As for our proposed HRAOD, it diverges from these traditional dehazing-dependent strategies but instead learns how to extract hazy features through distillation learning, thereby inherently strengthening its robustness to haze. Consequently, HRAOD achieves superior detection performance on real hazy aerial data.

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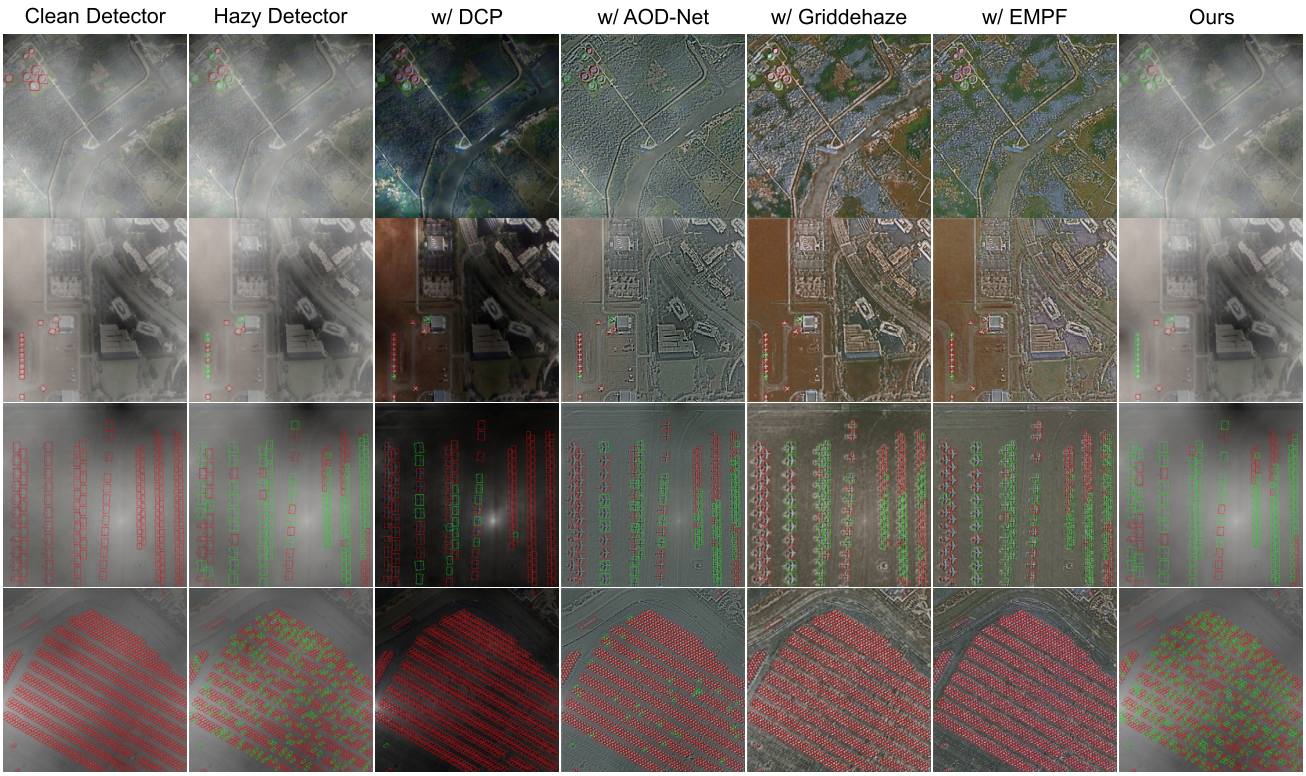



Fig. 6. Qualitative detection comparison on the SODA-A-Haze dataset. We use green, blue, and red boxes to denote true positive, false positive, and false negative predictions, respectively.

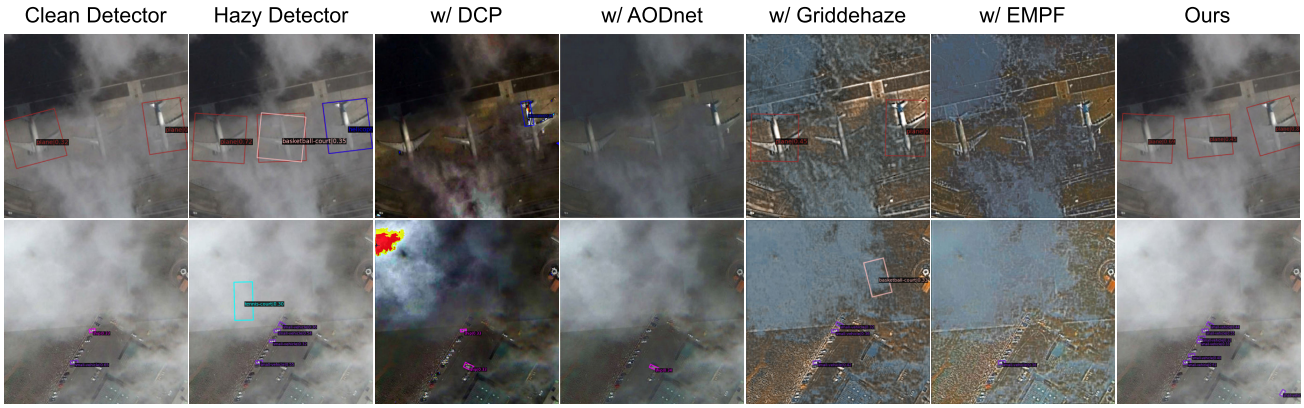



Fig. 7. Qualitative detection comparison on real-world hazy aerial scenes. Please zoom in for a better view.

### *F. Experimental Results on DroneVehicle Dataset*

In this section, we extend our experiments to drone aerial photography to verify the robustness and practicability of our method. The quantitative detection performance on 1073 real-world hazy images is reported in Table III. Since the real-world hazy images in DroneVehicle dataset are generally with light haze intensity, the detector trained on normal images can already achieve comparable detection performance. However, it still suffers a notable performance decrease in dominant class CA. For the detector trained on generated hazy images, although it achieves great improvement in class CA and FC, its performance in the other three classes has significantly declined. This decrease in performance is attributed to the degradation of image quality, which makes it more challenging for the detector to extract features and classify objects accurately. Methods with image dehazing as a pre-processing step do not stably improve the detection performance. For example, the method w/ DCP performs quite well in class CA but performs the worst in class BU. The method w/ GridDehaze achieves the best performance in VA class, but is worse than the other methods in terms of overall performance. This instability is mainly due to the objective mismatch between low-level and high-level vision tasks. Different from other methods, our HRAOD establishes a clean-to-hazy distillation framework, which can not only enhance the detector's robustness to hazy scenes but also retain the prior knowledge of the teacher detector for better localization and classification. Consequently, our method achieves the best detection performance in real-world hazy scenes and surpasses the second-best method by 3.9 points in mAP. We also demonstrate some visualization results in Fig. 8. Although the

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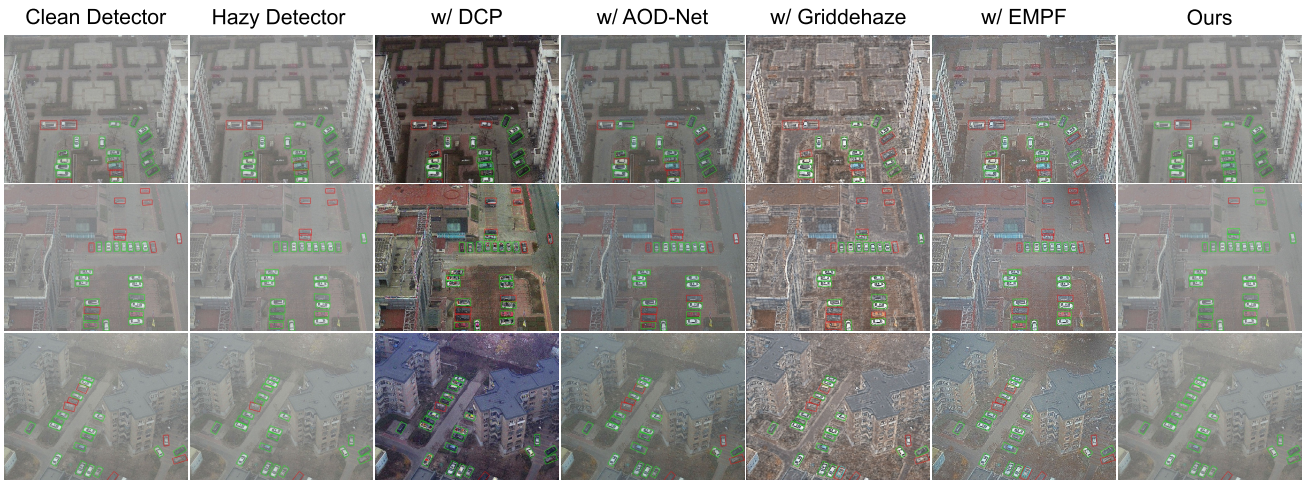



Fig. 8. Qualitative detection comparison on real-world hazy scenes from the DroneVehicle dataset. We use green, blue, and red boxes to denote true positive, false positive, and false negative predictions, respectively. Please zoom in for a better view.

TABLE III
QUANTITATIVE DETECTION COMPARISON ON THE REAL-WORLD HAZY SCENES FROM DRONEVEHICLE DATASET. BOLD TEXT SHOWS THE BEST RESULTS

| Method | CA | TR | BU | VA | FC | mAP |
|---|---|---|---|---|---|---|
| **Directly Detect:** | | | | | | |
| Clean Detector | 77.8 | 15.3 | 54.9 | 21.4 | 20.8 | 38.0 |
| Hazy Detector | 84.9 | 13.3 | 49.6 | 17.3 | 26.1 | 38.2 |
| **Dehaze + Detect:** | | | | | | |
| w/ DCP | 84.6 | 13.8 | 45.9 | 22.1 | 23.9 | 38.1 |
| w/ AOD-Net | 77.8 | 17.3 | 48.9 | 20.9 | 21.6 | 37.3 |
| w/ GridDehaze | 77.2 | 13.7 | 55.3 | **22.9** | 16.6 | 37.1 |
| w/ EMPF-Net | 77.5 | 15.6 | 53.5 | 21.0 | 19.0 | 37.3 |
| **Distillation:** | | | | | | |
| HRAOD (Ours) | **84.9** | **21.2** | **56.0** | 21.9 | **26.4** | **42.1** |

Clean Detector is trained on clean images, and directly tested on hazy images. Hazy Detector is trained and tested on hazy images.

TABLE IV
ABLATION STUDY ON EACH DISTILLATION STRATEGY

| $L_{grad}$ | $L_{degra}$ | mAP |
|---|---|---|
| | | 58.0 |
| ✓ | | 60.2 (+2.2) |
| | ✓ | 60.0 (+2.0) |
| ✓ | ✓ | **60.8 (+2.8)** |

traditional dehazing method DCP produces visually favorable results in real-world hazy scenes, it does not contribute significantly to the subsequent detection task. This may be because the DCP method can affect the appearance of the car (some white cars become colored after image dehazing). Among all competing methods, our HRAOD yields more correct predictions in real-world hazy scenes, which further demonstrates its superiority.

### G. Ablation Study

*1) Break-Down Ablation:* In the HRAOD, we first construct a clean-to-hazy distillation paradigm and then propose two distillation strategies to promote knowledge transfer. We perform a break-down ablation study to analyze their influence, and the experimental results are provided in Table IV. It can be seen that the gradient-guided feature imitation strategy $L_{\text{grad}}$ improves the baseline model by 2.2 points on mAP, and the degradation-weighted response distillation $L_{\text{degra}}$ improves the baseline by 2.0 points. When simultaneously distilling the feature and response information, our HRAOD leads to an improvement of 2.8 points on mAP. These results verify the importance of each component of our methods.

*2) Ablation Study on Gradient Distillation:* In our HRAOD, we propose to minimize the discrepancy between the gradients of the student feature maps and those of the teacher feature maps, aiming to enhance image contrast and improve the student network's ability to extract valuable features in hazy conditions. To further validate the efficacy of this approach, we replaced the gradient distillation strategy with the classical feature distillation strategy, denoted as "+ feat imitation" in Table V. As demonstrated in the results, introducing feature imitation resulted in a decrease of 1.0 points on mAP. This is primarily due to the significant domain gap between clean and hazy features. Directly minimizing the distance between these features makes it difficult for the student network to grasp the learning focus. Whereas traditional feature distillation methods fail to convey information from clean images effectively, our proposed gradient distillation approach encourages the detector to increase the contrast of hazy features, thereby preserving more semantic information. To intuitively demonstrate the contribution of gradient distillation, we further visualize the bottom-layer feature maps of different models in Fig. 9. It can be observed that, due to the interference of haze, the baseline model mistakenly activates some redundant background regions. Due to the excessive gap between clean features and hazy features, directly introducing classical feature distillation makes it difficult for the network to grasp the focus of learning, thus failing to effectively extract valuable features of the desired objects. In contrast, our approach strengthens

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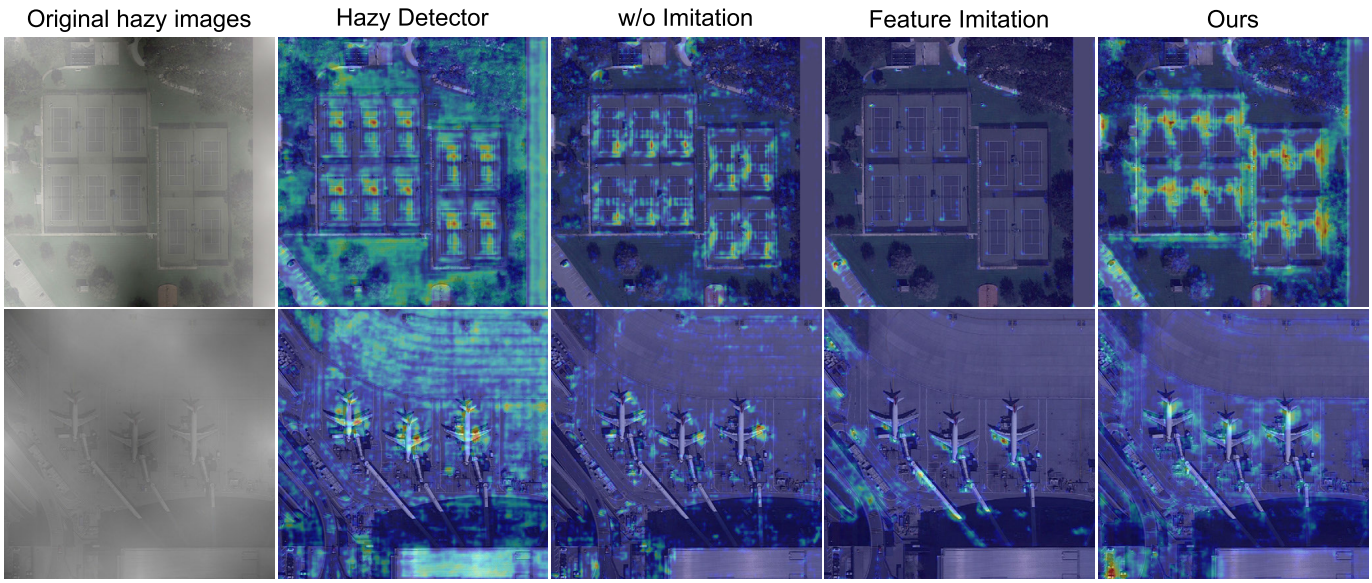



Fig. 9. Activation feature map of different ablation models and our HRAOD. Our method can accurately grasp the representative features of desired objects and surpass background noise under hazy conditions.

TABLE V
ABLATION STUDY ON GRADIENT IMITATION STRATEGY

| w/o Imitation | + Feat Imitation | HRAOD |
|---|---|---|
| 60.0 | 59.0 (-1.0) | **60.8 (+0.8)** |

TABLE VI
ABLATION STUDY ON THE DISTILLATION REGION

| Baseline | GT Regions | Equal Weights | Degradation Map |
|---|---|---|---|
| 58.0 | 58.3 (+0.3) | 59.4 (+1.4) | **60.0 (+2.0)** |

the edge information of objects through gradient distillation, thereby suppressing the activation of the background part and concentrating on information extraction of objects and their surroundings.

*3) Ablation Study on Response Distillation Region:* In our HRAOD, we design a degradation-weighted response distillation to enhance the prediction robustness of severely degraded regions. To verify its contribution, we further introduce two variant models with different response distillation strategies. As demonstrated in Table VI, only distilling the positive sample region (denoted as "GT Regions") produces an improvement of 0.3 points in mAP, which is because some valuable information outside the GT boxes still deserve transferring [41]. We further distill all regions of the response map, but with equal weights for each pixel (denoted as "equal weights"), which brings a 1.4-points improvement compared with the baseline. Although this distillation strategy can transfer information from every region, it treats all areas equally, ignoring the nonuniform distribution of haze. For objects degraded by light haze, detectors can adapt to these data from the training process on the hazy domain. However, it is challenging for detectors to achieve accurate detection results on heavily degraded objects, which is the focus of clean-to-hazy distillation. With the proposed degradation-weighted distillation strategy, our HRAOD can selectively distill the response information according to the hazy degree, thereby enhancing the detector's ability to detect heavily degraded objects.

## V. DISCUSSION

### A. Impact of Haze on Different Object Categories

Based on the experimental results from the DOTA-Haze and SODA-A-Haze datasets, we observed significant variations in the performance drop among different object categories in the presence of haze. For example, compared with our HRAOD, the hazy detector in Table I can already achieve satisfactory performance on PL (1.1% mAP drop) and ST (with the same performance), and it can achieve comparable or even slightly higher performance on AP and WM categories in Table II. In contrast, certain objects such as GTF (with 20.7% drop in mAP) in Table I and CO (with an 18.1% drop) in Table II suffer from severe performance degradation. It is interesting to investigate the reason behind this disparity.

After analyzing the degradation pattern of haze and the characteristics of the aforementioned objects, we posit that the color of objects significantly contributes to this observed disparity. As demonstrated in Fig. 10, objects such as the white plane and ST tend to retain more original features as their inherent color closely resembles that of haze, allowing them to maintain visibility. In contrast, objects with distinct colors, such as the GTF with deep green and colorful COs, undergo substantial color shifts, which may compromise their unique features for recognition. Compared with the baseline model, our proposed HRAOD leverages a degradation-weighted response distillation scheme to promote the prediction information transfer in severely degraded regions. For objects with haze-like colors, the baseline model can already satisfactorily detect them due to their well-preserved features. Thus,

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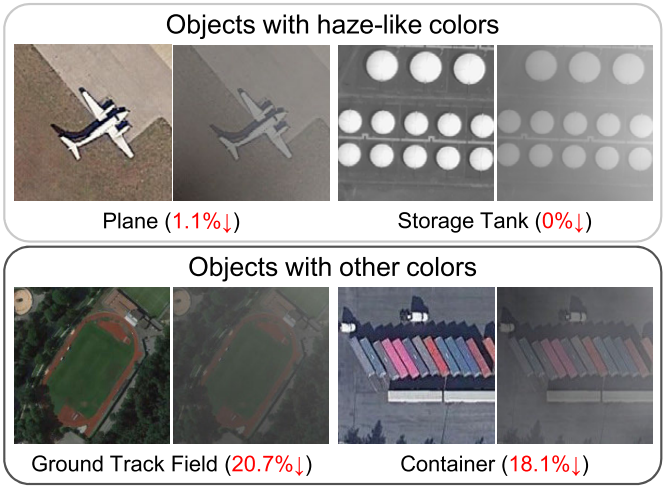



Fig. 10. Clean-hazy image patch pairs of some representative categories. We use the red font to present the performance gap in mAP between the hazy detector and our proposed HRAOD.

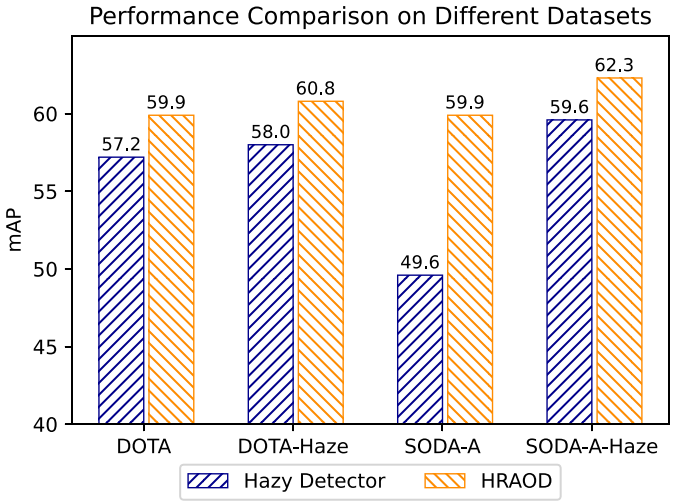



Fig. 11. Detection performance comparison between hazy detector and our proposed HRAOD on both the clean and hazy dataset.

HRAOD assigns less weight to them during the response distillation. Conversely, for those objects experiencing significant color shifts under hazy conditions, HRAOD pays more attention to aligning their predictions more closely with those derived from clear conditions, thereby substantially improving the detector's robustness to haze.

### B. Detection Performance on Clean Images

So far, we have presented a robust detector for hazy scenes. Given the varied conditions encountered in real-world scenarios, it is also necessary to evaluate how the hazy-robust detectors perform on clean aerial scenes. We further compare the performance of the hazy detector and our HRAOD on the clean dataset (DOTA and SODA-A). As demonstrated in Fig. 11, our proposed HRAOD outperforms hazy detector comprehensively on both clean and hazy datasets. Although directly optimizing detectors on hazy images can increase detection robustness, their performance on clean images is greatly compromised due to the gap between clean and hazy datasets. Moreover, this method is still hindered by the low contrast and blurred edges caused by haze, resulting in ineffective feature extraction and inaccurate prediction. Conversely, during the distillation process of our HRAOD, we leverage information from both clean and hazy datasets. The student model naturally inherits the general representation learned by the teacher detector, which is also valuable for detection on clean images. As a result, our method not only excels in detecting objects under hazy conditions but also maintains the detection performance on clean images.

TABLE VII
MAP UNDER DIFFERENT HAZE INTENSITY ON THE DOTA-HAZE DATASET. BOLD TEXT SHOWS THE BEST RESULTS

| Method | Light | Moderate | Heavy |
|---|---|---|---|
| Clean Detector | 47.5 | 39.4 | 25.6 |
| Hazy Detector | 59.5 | 57.2 | 55.2 |
| w/ DCP | 48.8 | 45.4 | 34.7 |
| w/ AOD-Net | 39.9 | 33.5 | 19.7 |
| w/ GridDehaze | 50.1 | 43.6 | 34.3 |
| w/ EMPF-Net | 49.5 | 43.9 | 30.5 |
| HRAOD(ours) | **62.3** | **60.3** | **58.7** |

### C. Effect of Haze Intensity

In this section, we delve deeper into the impact of haze intensity on detection performance. Specifically, we use the ImageNet-C benchmark to synthesize aerial images with varying degrees of haze intensity. We choose the DOTA dataset for experiment and curated three datasets with different levels of haze intensities, namely light, moderate, and heavy. The detection performance of different methods is demonstrated in Table VII. As can be seen from the results, although the performance of all methods decreases as the haze intensity increases, the robustness of HRAOD against different hazy intensity prevails among the competing methods. While the performances of best image dehazing methods, DCP and GridDehaze, degrade over 14 points in mAP from light haze to heavy haze, our method only shows a 3.6-point dip, which further demonstrates the robustness of our method. It is also worth noting that although the model w/ EMPF-Net achieves better performance than the model w/ DCP under light haze, it falls behind w/ DCP under heavy haze. This experiment result demonstrates that the image dehazing method might not stably contribute to object detection.

## VI. CONCLUSION

In this article, we present HRAOD, a knowledge distillation framework designed for boosting detection robustness in hazy aerial scenes. HRAOD adopts a clean-to-hazy information distillation paradigm, directly empowering the detector to "see through haze" without involving the image dehazing process. Considering the degradation pattern of haze, we further introduce two distillation strategies to promote information transfer. Firstly, a gradient-guided feature imitation strategy is employed to enhance the quality of hazy features, alleviating the blurred boundaries and low contrast induced by haze.

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Secondly, a degradation-weighted response distillation strategy is developed to adaptively narrow the distance between the response maps of student and teacher models, leading to more precise localization and classification outputs. Due to the lack of available detection-oriented hazy aerial data, we contribute three hazy aerial image datasets and conduct extensive experiments on them. Experiment results on both simulation and real-world hazy images demonstrate that our method achieves significant detection improvements under hazy conditions compared to the baseline model and the mainstream "dehaze + detect" models. Moreover, as no additional image dehazing step is required, our method enhances detection robustness against hazy conditions without incurring additional inference costs. Considering the limitations imposed by data scarcity on the research development of aerial object detection in adverse conditions, we plan to collect and annotate more real-world aerial hazy images in the future and establish a benchmark. We also plan to investigate how other degradation factors affect aerial object detection and their corresponding solution.

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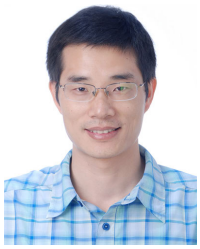
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